Homework #1

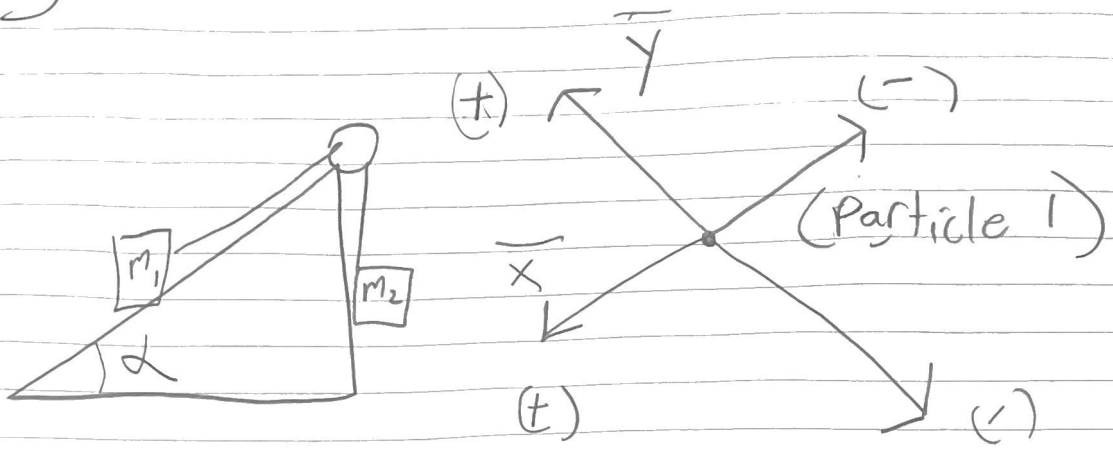
Parker Lewis

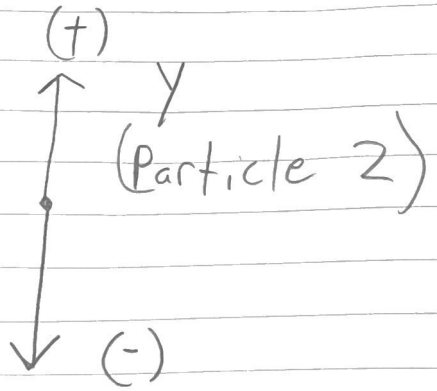
prob 1.)

Time	spent
1 (a)	1
(b)	2
(c)	1
Prep	2

a.)

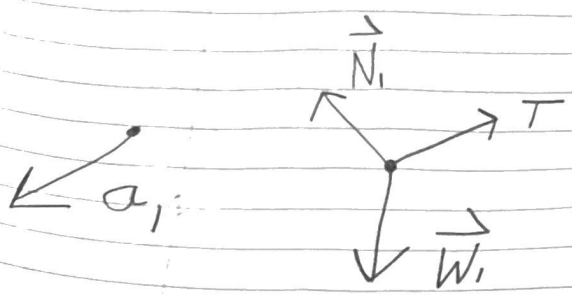
Part a.)



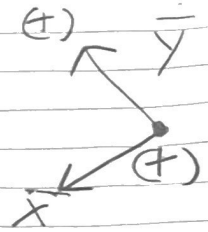


part b.)

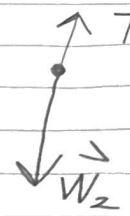
m_2 going up:



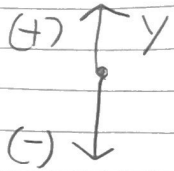
(particle 1)



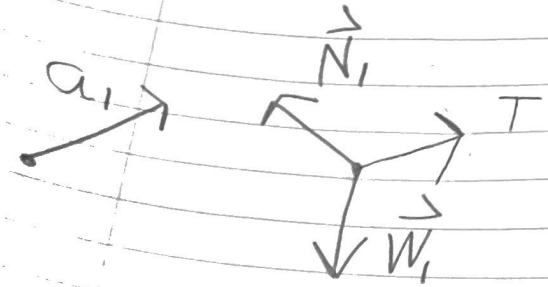




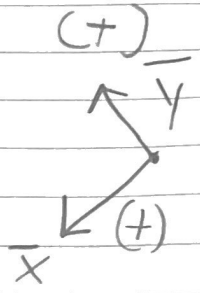
(particle 2)

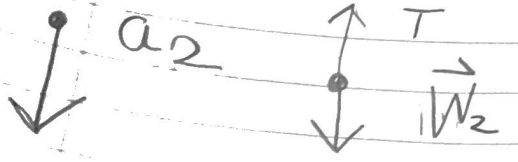


m_2 going down:

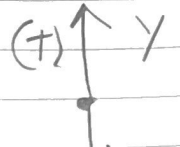


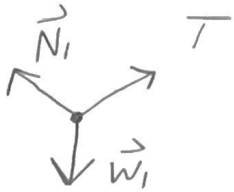
(particle 1)





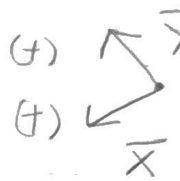
(particle 2)





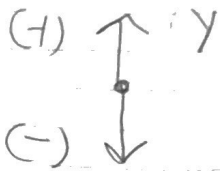
At Rest:

(particle 1)



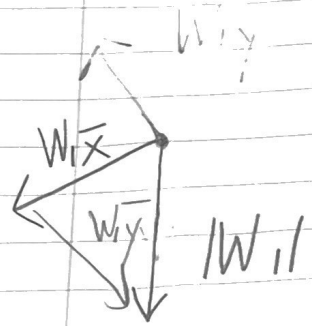


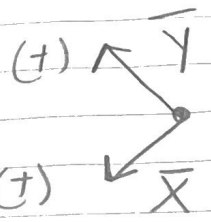
(particle 2)

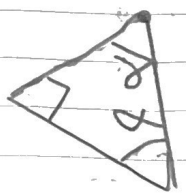


particle c.)

(particle 1)







$$\gamma = 90 - \alpha$$

part d.)

M2 sliding up:

$$x_1 = x_2 + l$$

$$v_1 = v_2$$

$$a_1 = a_2$$

$a_1 = a_2 = a$

$\sum F_{\bar{x}} = m_1 a$

$\sum F_{\bar{y}} = 0$

$\sum F_y = m_2 a$

m_2 sliding down:

$\sum F_{\bar{x}} = m_1 a$

$\sum F_{\bar{y}} = 0$

$\sum F_y = m_2 a$

At Rest:

$\sum F_{\bar{x}} = 0$

$\sum F_{\bar{y}} = 0$

$\sum F_y = 0$

m_2 sliding up:

$$\sum F_{\bar{x}} = m_1 a$$

$$m_1 g \sin\alpha - T = m_1 a$$

$$\sum F_{\bar{y}} = 0$$

$$|N_1| - m_1 g \cos\alpha = 0$$

$$\sum F_y = m_2 a$$

$$T - m_2 g = m_2 a$$

m_2 sliding down:

$$\sum F_{\bar{x}} = -m_1 a$$

$$-m_1 g \sin\alpha + T = -m_1 a$$

$$\sum F_{\bar{y}} = 0 \qquad |N_1| - m_1 g \cos\alpha = 0$$

$$\sum F_y = -m_2 a, \quad m_2 - T = -m_2 a$$

Rest:

$$\sum F_x = 0$$

$$m_1 g \sin\alpha - T = 0$$

$$\sum F_y = 0$$

$$|N_1| - m_1 g \cos\alpha = 0$$

$$\sum F_y = 0$$

$$T - m_2 g = 0$$

part e.)

M_2 sliding up:

$$\boxed{|N_1| = m_1 g \cos\alpha}$$

$$m_1 g \sin\alpha - T = m_1 a$$

$$T - m_2 g = m_2 a$$

$$-T + m_1 g \sin\alpha = m_1 a$$
$$T - m_2 g = m_2 a$$

$$m_1 g \sin\alpha - m_2 g = (m_1 + m_2) a$$

$$\frac{g(m_1 \sin\alpha - m_2)}{(m_1 + m_2)} = a = a_1 = a_2$$

$$-T + m_1 g \sin\alpha = m_1 a \quad (1)$$
$$T - m_2 g = m_2 a \quad (2)$$

* multiply m_2 to (1). m_1 to (2)

$$-m_2 T + m_1 m_2 g \sin\alpha = m_1 m_2 a$$
$$m_1 T - m_2 m_1 g = m_1 m_2 a$$

* subtract (1) to (2)

$$-T(m_2 + m_1) + g m_1 m_2 (\sin\alpha + 1) = 0$$

$$M = \frac{m_1 m_2}{(m_1 + m_2)}$$

$$T = gM(\sin\alpha + 1)$$

$\vec{W}_1 = m_1 g \sin\alpha \, \hat{x} - m_1 g \cos\alpha \, \hat{y}$

$\vec{W}_2 = -m_2 g \, \hat{y}$

$\vec{N}_1 = m_1 g \cos\alpha \, \hat{y}$

$|\vec{W}_1| = m_1 g$

$|\vec{W}_2| = m_2 g$

m_2 sliding down:

- mass values are unknown
- same magnitude values of forces will be same but directions are different by applying 3rd Law ($\vec{F}_{ij} = -\vec{F}_{ij}$)

At Rest :

$$\sum F_x = 0$$

$$m_1 g \sin\alpha - T = 0$$

$$T = m_1 g \sin\alpha$$

$$\sum F_y = 0 \qquad \boxed{|\vec{N}_1| = m_1 g \cos\alpha}$$

$$\boxed{a_1 = a_2 = 0 = a}$$

$$\sum F_y = 0 \qquad T - m_2 g = 0$$

$$\boxed{T = m_2 g = m_1 g \sin\alpha}$$

$$\text{if } \frac{m_2}{m_1} = \sin\alpha$$

$$\vec{N}_1 = m_1 g \cos\alpha \, \hat{y}$$

$$\vec{w}_1 = m_1 g \sin\alpha \, \hat{x} - m_1 g \cos\alpha \, \hat{y}$$

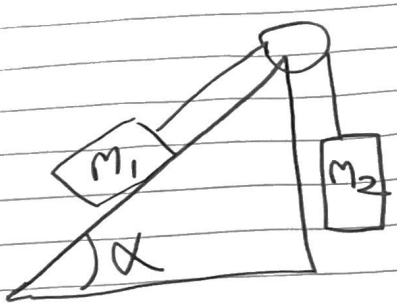
$$\vec{w}_2 = -m_2 g \, \hat{y} \quad , \quad |\vec{w}_1| = m_1 g \quad |\vec{w}_2| = m_2 g$$

b.)

Case #1)

At Rest:

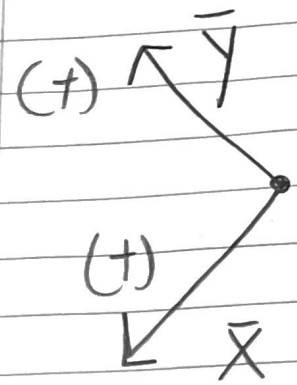
a.)

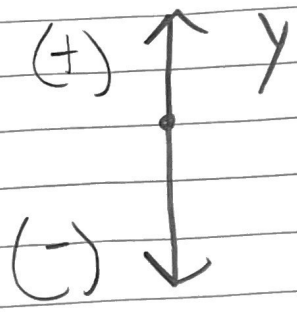


$$x_1 = x_2 + \delta$$

$$\dot{x}_1 = \dot{x}_2$$

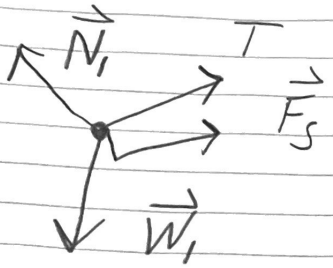
$$a_1 = a_2 = a$$

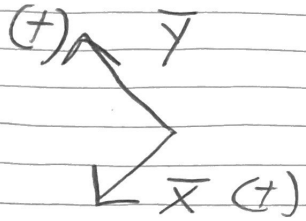




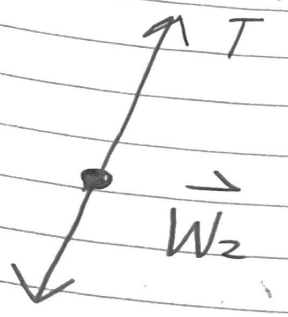
b.)

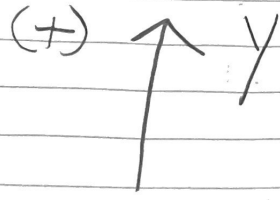
Particle 1 :





Particle 2 :



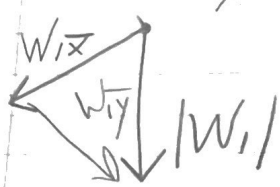


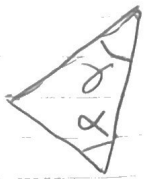
$a = 0$

(systems at rest)

c.)

$\gamma = 90 - \alpha$





d.)

$$\sum F_x = 0$$

$$m_1 g \sin\alpha - T - |F_s| = 0$$

$$\sum F_y = 0$$

$$|N_1| - m_1 g \cos\alpha = 0$$

$$\sum F_y = 0 \qquad T - m_2 g = 0$$

e.)

$$m_1 g \sin\alpha - T - |F_s| = 0$$

$$0 \leq |F_s| \leq \mu_s |N_1|$$

$$T - m_2 g = 0$$

$$T = m_2 g$$

$$m_1 g \sin\alpha - m_2 g - |F_s| = 0$$

$$|F_s| = m_1 g \sin\alpha - m_2 g$$

$$g(m_1 \sin\alpha - m_2) \leq \mu_s m_1 g \cos\alpha$$

$$m_1 \sin\alpha - m_2 \leq \mu_s m_1 \cos\alpha$$

$$m_1 \sin\alpha \leq \mu_s m_1 \cos\alpha + m_2$$

Condition for system to stay at rest

$$T = m_2 g = m_1 g \sin\alpha - |F_s|$$

$$|\vec{N}_1| = m_1 g \cos\alpha$$

$$|F_s| \leq \mu_s m_1 g \cos\alpha$$

$$\vec{N} = m_1 g \cos\alpha \, \hat{y}$$

$$\vec{w}_1 = m_1 g \sin\alpha \, \hat{x} - m_1 g \cos \hat{y}$$

$$|\vec{w}_1| = m_1 g$$

$$\vec{w}_2 = -m_2 g$$

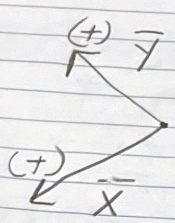
$$|\vec{w}_2| = m_2 g$$

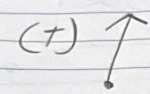
$$\vec{F}_s = -f_{req} \, \hat{x}$$

$$a = 0$$

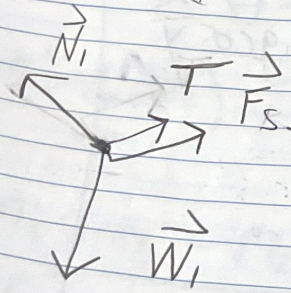
Case #2 :

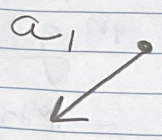
Box 2 moves up :



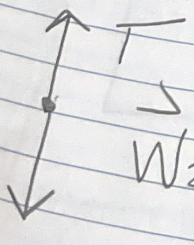


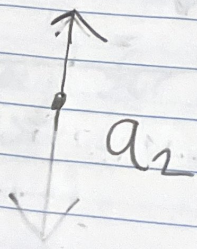
(particle #1)

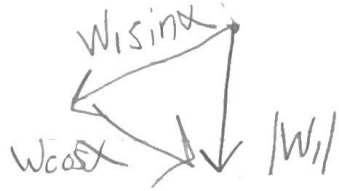




(Particle #2)







$$\left(\begin{array}{l} a_1 = a_2 = a \\ \text{from} \quad x_1 = x_2 + \ell \end{array} \right)$$

$$\sum F\bar{x} = m_1 a$$

$$m_1 g \sin\alpha - T - |\vec{F}_s| = m_1 a$$

$$|F_s| \leq \mu_s |N_1|$$

$$\sum F\bar{y} = 0$$

$$\boxed{|N_1| = m_1 g \cos\alpha}$$

$$\sum F_y = m_2 a$$

$$T - m_2 g = m_2 a$$

$T - m_2 g = m_2 a$

$-T + m_1 g \sin\alpha - |\vec{F_s}| = m_1 a$

$-m_2 g + m_1 g \sin\alpha - |\vec{F_s}| = a(m_1 + m_2)$

$|\vec{F_s}| \leq m_1 g \cos\alpha \, \mu_s$

$|\vec{F_s}| = -m_2 g + m_1 g \sin\alpha - a(m_1 + m_2)$

$\mu_s m_1 g \cos\alpha \leq m_1 g \sin\alpha - m_2 g - a(m_1 + m_2)$

To start moving up

$|\vec{F_s}| \leq m_1 g \cos\alpha \, \mu_s$ $\quad$ $|\vec{F_s}| = f_{req}$

$f_{req} = m_1 g \sin\alpha - m_1 g - a(m_1 + m_2)$

$$\frac{f_{req} + m_2 g - m_1 g \sin\alpha}{(m_2 + m_1)} = a$$

$$\vec{W}_1 = m_1 g \sin\alpha \,\hat{x} - m_1 g \cos\alpha \,\hat{y}$$

$$|\vec{W}_1| = m_1 g$$

$$\vec{N}_1 = m_1 g \cos\alpha \,\hat{y}$$

$$\vec{F}_S = -f_{rg}\,\hat{x}$$

$$T - m_2 g = m_2 a$$

$$-T + m_1 g \sin\alpha - f_{rg} = m_1 a$$

$$m_1 T - m_1 m_2 g = m_1 m_2 a$$

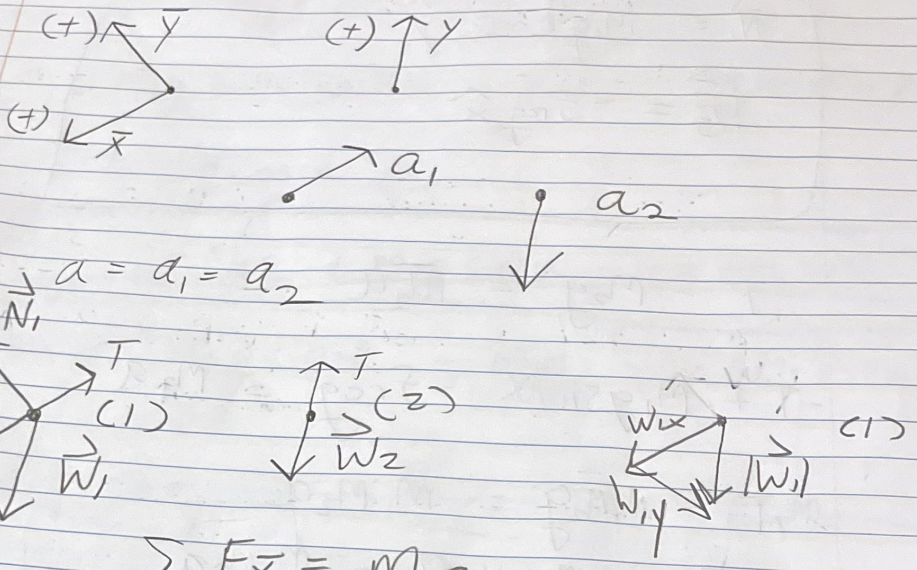
$$-m_2 T + m_2 m_1 g \sin\alpha = m_1 m_2 a$$

$$m_2 T + m_1 T - m_1 m_2 g - m_1 m_2 g \sin\alpha = 0$$

$$T(m_1 + m_2) = m_1 m_2 g + m_1 m_2 g \sin\alpha$$

$$T = \frac{m_1 m_2 g + m_1 m_2 g \sin\alpha}{(m_1 + m_2)}$$

Case #3 : moving down



$$\sum F_{\vec{x}} = m_1 a$$

$$|\vec{F}_s| = f_{reg} \qquad |\vec{F}_s| \leq \mu_s |\vec{N}_1|$$

$$-|\vec{F}_s| \quad T - m_1 g \sin\alpha = -ma$$

$$\sum F_{\vec{y}} = m_2 a$$

$$-T + m_2 g = -m_2 a$$

$$\sum F_y = 0, \qquad |\vec{N}_1| = m_1 g \cos\alpha$$

$$|\vec{F_s}| \leq \mu_s m_1 g \cos\alpha$$

$$\begin{cases} -|\vec{F_s}| + T - m_1 g \sin\alpha = -m_1 a \\ m_2 g - T = -m_2 a \end{cases}$$

$$-|\vec{F_s}| + m_2 g - m_1 g \sin\alpha = -a(m_1 + m_2)$$

$$m_2 g - m_1 g \sin\alpha + a(m_1 + m_2) = |\vec{F_s}|$$

$$\mu_s m_1 g \cos\alpha \leq m_2 g - m_1 g \sin\alpha + a(m_1 + m_2)$$

(start moving down)

$$\therefore f_{req} = m_2 g - m_1 g \sin\alpha + a(m_1 + m_2)$$

$$\frac{f_{req} + m_1 g \sin\alpha - m_2 g}{(m_1 + m_2)} = a$$

$$T = -m_2 a - m_2 g$$

$$\boxed{T = -m_2 g - m_2 \left(\frac{f_{reg} + m_1 g \sin\alpha + m_2 g}{m_1 + m_2}\right)}$$

$|\vec{W}_1| = m_1 g$

$|\vec{W}_2| = m_2 g$

$\vec{W}_2 = -m_2 g \hat{y}$

$\vec{W}_1 = m_1 g \sin\alpha \hat{x} - m_1 g \sin\alpha \hat{y}$

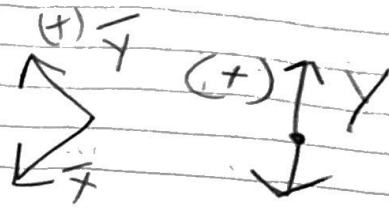
$\vec{F}_s = f_{reg} \hat{x}$

$|\vec{F}_s| = f_{reg}$

c.)

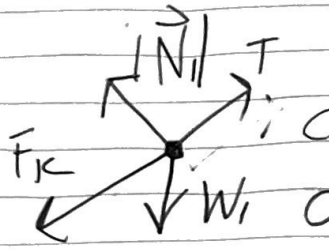
Moving down case:

Case #1:

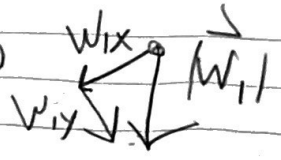


$$a = \frac{\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g}{(m_1 + m_2)}$$

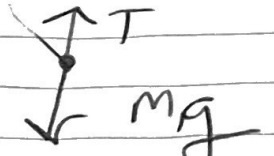
$$F_k = \mu_k m_1 g \cos\alpha$$



(1) $a_1 = a_2 = a$



(2)





(2)

$$a = \frac{\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g}{(m_1 + m_2)}$$

$a \leq 0$ (stopping)

$$\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g \leq 0$$

$$m_2 \geq \mu_k \cos\alpha\, m_1 + m_1 \sin\alpha$$

stopping

$$V_f = V_0 - at$$

($V_f = 0$, time at stop)

$$0 = V_0 - at$$

$$\frac{-V_0}{-a} = t$$

$$t = \frac{V_0(m_1 + m_2)}{(\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g)}$$

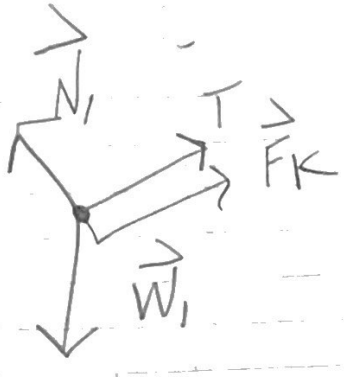
$$\Delta X = V_0 t - \frac{1}{2} a t^2$$

$$\Delta X = \frac{V_0^2(m_1+m_2)}{(\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g)} \cdot \left(V_0(m_1\right.$$

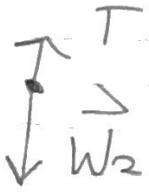
$$-\frac{1}{2}\left(\frac{V_0^2(m_1+m_2)}{(\mu_k m_1 g \cos\alpha + m_1 g \sin\alpha - m_2 g)}\right)$$

case #2

moving up:



(1)

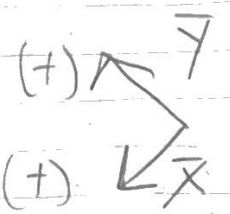


(2)

$a_1 = a_2 = a$









$a \geq 0$ (increase speed)

Previous parts:

$$\frac{M_K m_1 g \cos\alpha + m_2 g - m_1 g \sin\alpha}{(m_1 + m_2)} = a$$

$$M_K m_1 g \cos\alpha + m_2 g - m_1 g \sin\alpha \geq 0$$

$$m_1 \sin\alpha \leq M_K m_1 \cos\alpha + m_2$$

Since the system is coupled

$(a_1 = a_2)$

For box 2 to move with constant speed ($a = 0$).

which means that the system would be at rest.

In conclusion. $a = 0$ only if the system is at rest. Moving at constant speed still means ΣF of motion of system that experiences forces.